



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

PROGRAMME TITLE: Bachelor of Information and Communications Technology

PROGRAMME CODE: BINCT1

Study Guide

2026

MODULE: Integrative Programming and Technologies 3

MODULE CODE: IPRT301

SAQA CREDITS: 16

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Lectures: Three (3) theory periods per week (with lecturer)
 Practical/Tutorials: An additional practical period per week (at the lab)
 Lecture venue: See the approved timetable
 Practical/Tutorial venue: See the approved timetable
 Duration: 16 weeks
 Relevant Policies and rules: See Departmental Handbook.

1. Welcome

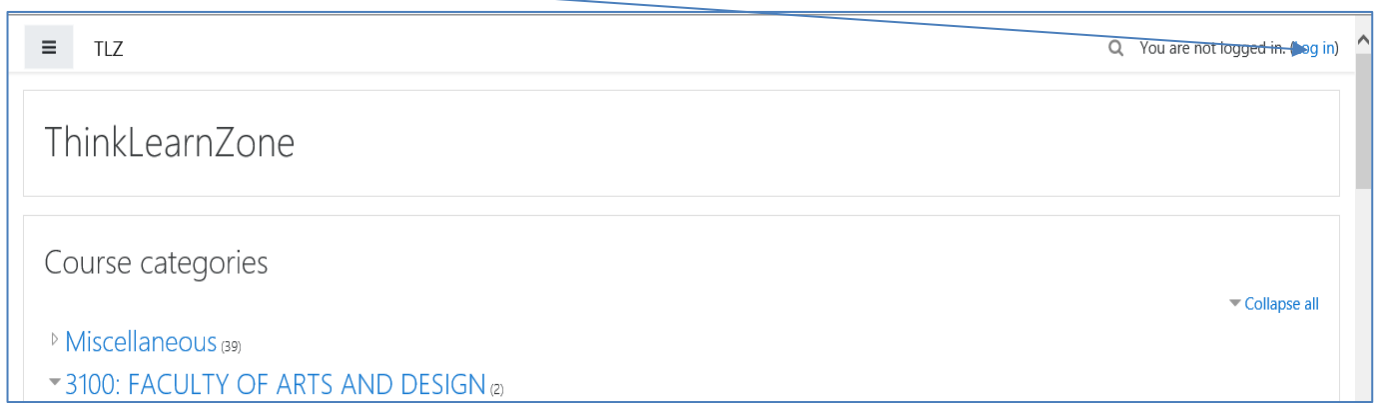
Welcome to Integrative Programming and Technologies. The purpose of this learner guide is to provide important information on some aspects of the subject and to orientate the learner.

This guide contains relevant information to help you prepare for the module. This includes: the lecturer; expected learning outcomes, assessment strategies and criteria; and schedules for teaching and assessments.

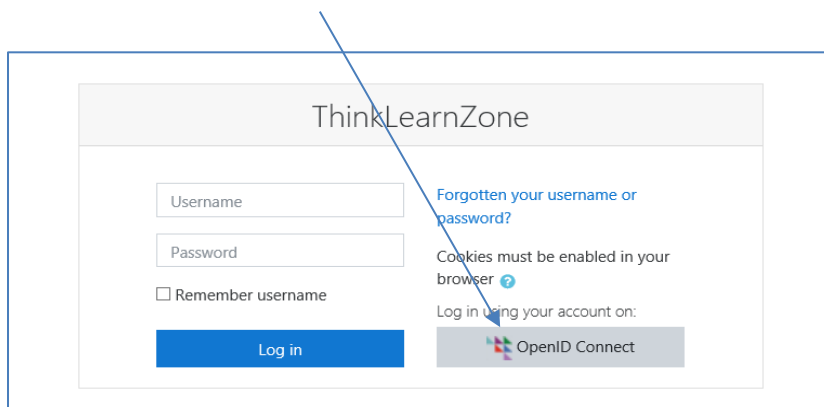
2. Using your online ThinkLearnZone (Moodle) classroom

All taught subjects/modules have their own online classroom on the ThinkLearnZone. You can access your classroom at <https://tlzprod.dut.ac.za>

Click on (Log in) on the top right corner



On the next screen click on “OpenID Connect”



Use your dut4life email address to sign (yourstudentnumber@dut4life.ac.za) and your email password as your password (Dut + first six digits of your ID no., e.g. Dut980610. For first-time 2020 students - use \$\$Dut + first six digits of your ID no., e.g. \$\$Dut980610)

Ask your lecturer for guidance or check out “how to log in” from here:

<https://www.dut.ac.za/faq-students/>

The e-learning website also has contact information for help and technical assistance <https://www.dut.ac.za/elearning-contacts/>

You can call the e-learning helpdesk on 031 373 67810 or email them on tlzsupport@dut.ac.za

3. Introduction to the Module

This module focuses on the integration of organizational resources (e.g., data; software and hardware; and other platforms) and attendant issues, e.g., communication, standardization, security, etc. On completion, learners will be able to analyze business problems; select the appropriate techniques and technologies; and integrate heterogeneous resources to achieve organizational objectives.

4. Learning outcomes and assessment criteria

The table below shows the links between the module outcomes and assessment criteria and methods.

Learning outcomes	Assessment criteria	Assessment methods
Describe and contrast the different integrating systems architectures	Understand the different integration architectures and know which to select to solve specific problems	The learning outcomes will be assessed using various methods such as the following: Quizzes Tests Assignments Project
Design, develop and test a program that applies two or more design patterns to solve a problem	Apply design patterns and best practices to create and code classes, inheritance and interfaces	
Design, develop and test a programs that use XML technologies	Write programs that use the XML technologies to exchange data	
Design, develop and test a program that uses a messaging service	Write programs that allows synchronous and asynchronous messaging between applications	
Write, debug and test an interactive web-based application that uses server-side script to process input from a web page Perform a security audit of the c of an existing system	Write a web application that uses server-side scripts to process request Apply security audit practices to analysing code	

5. Learning, teaching and assessment strategies

This module adopts an integrated teaching and learning approach. Learning is facilitated by face-to-face contact, self-learning and research. The following is an outline of the strategies.

5.1 Learning activities

There will be three formal lecture periods per week in which the basic concepts will be explained. In addition there will be one practical/tutorial period in which you will be required to practice (do homework, assignments, etc.). The following is a breakdown of learning activities and average learning times.

Learning activity	% learning time
Contact (Lectures)	40
Practical	20
Other assignments	20
Independent study	20

5.2 Graduate attributes

5.2.1. Knowledge of (designing and implementation) distributed software.

5.2.2. Critical thinking skills.

5.2.3. Problem solving skills

5.3 Assessment

The learning outcomes will be assessed as shown in the table below. Further information will be provided during lectures and/or on Moodle.

Assessment	%
Tests (and any other assessments)	40
Examination	60

5.3.1 Schedule of assessments

The table below shows the schedules of the main assessments. Note: these are subject to changes; which would be available on Moodle

Assessment	Week
Test 1	16 and 23 March 2026
Test 2	27 April and 4 May 2026
Test 3 (Make-up)	Week of 11 May 2026

5.3.2 Requirements for Assessments

All submissions (e.g., assignments and project) should have a cover page with the following details: Student Name, Student Number, Module Name, Program Name, Assignment Number and Assignment Title. If it is a group assignment the details of all group members must appear on the cover.

5.3.3 Due Dates

Due dates for all work, **must** be adhered to. If you default you will be **penalized 5% for every day (weekends inclusive)** that the work is late. If a valid reason for late submission is given in writing, then the penalty will be left totally to the discretion of the lecturer.

5.3.4 Absence from a Test

If a student is absent for a test, it is his/her responsibility to provide the subject lecturer with an explanation, together with any documentation in support thereof, **not later than a week after the date of the scheduled test**. Students that are granted permission to write a make-up test will be expected to cover the entire syllabus content for the test. The make-up test will most likely be scheduled for the latter part of the semester. Announcements will be made in lectures and on Moodle. It is the students responsibility to liaise with the lecturer to confirm whether they qualify to write the make-up test.

5.4 Activities to promote learning

These include:

- 5.4.1. Individual and group research
- 5.4.2. Hands on learning (from designing to developing a solution for a real world problem)
- 5.4.3 Resources (e.g., copies of lecture notes, tutorials questions and solutions, etc.), may be posted on Moodle (and, where applicable, handed out). In addition, open source ebooks and other online resources may be provided to supplement teaching and learning.

5.5 Library orientation

The Library provides orientation to all students as part of the orientation process, and may have adequately covered this in the previous years. You are, therefore, expected to be familiar with the library and how to acquire resources. The subject librarians are valuable sources of information.

6. Copyright and plagiarism

According to, the Merriam-Webster online dictionary, to "plagiarize" means:

- "to steal and pass off (the ideas or words of another) as one's own
- to use (another's production) without crediting the source
- to commit literary theft
- to present as new and original an idea or product derived from an existing source "

The university, faculty and department emphasizes the need for academic integrity in all student learning and assessment activities. Various workshops and training sessions are conducted in the course of the programme to inform and train students on the plagiarism policies, procedures and guidelines as well as the relevant tools and techniques for referencing. In particular, the DUT plagiarism declaration (see Appednxi A) must be completed and submitted with assignments, portfolios and other formative assessments.

7. Scheme of work

The table below shows the scheme of work.

Week	Commences	Topic	Deliverable
1	09-Feb-26	Overview: orientation	
		Overview: Development environment	Install JDK; install and configure IDE (including plugins) Install and configure Tomcat (later in the course)
		Overview: Integrative programming and Technologies	
		Overview: Java programming	Write simple Java class and Tester
2	16-Feb-26	Integrative Coding	Apply inheritance, interfaces and patterns in Java applications
3	23-Feb-26	Integrative Coding	
4	02-Mar-26	Integrative Coding	
5	09-Mar-26	Data Mapping and Exchange /	Design and develop a simple Java/XML application and Java/XML web application
6	16-Mar-26	Test 1	Design and develop a simple Java/XML application and Java/XML web application
7	23-Mar-26	Test 2	Design and develop a simple Java/XML application and Java/XML web application
8	30-Mar-26	Vacation	
9	06-Apr-26	Data Mapping and Exchange	Design and develop a simple Java/XML application and Java/XML web application
10	13-Apr-26	Intersystem Communications	Design and develop simple distributed applications: sockets, URL and RMI
11	20-Apr-26	Intersystem Communications	Design and develop simple distributed applications: sockets, URL and RMI
12	27-Apr-26	Test 2/ Scripting techniques	Integrate scripts into an application
13	04-May-26	Test 2/Scripting techniques	Integrate scripts into an application
14	11-May-26	Make up (Test 3);	
15	18-May-26	END OF LECTURES	
16	25-May-26		
17	01-Jun-26		
18	08-Jun-26		Exam starts 10 June 2026
Note: Self study		Software Security Practices	Perform security audit Apply security best practices

8. Student Support

Should you require further support relating to the subject content, schedule a consultation with your lecturer within the allocated consultation times. Your lecturer may refer you to one of the support services available. You will be able to find a full list of services and comprehensive details for these services on the DUT website http://www.dut.ac.za/student_portal/. Find below a summarised list of the important support services available to you as students of DUT.

Support Service	Contact Details	What they can help with
Faculty Office	East Wing, Hotel School Building, Ritson Campus Ms Debbie Small (Faculty Assistant) Tel: (031) 373 5418 Email: deborahs@dut.ac.za	Registration, remark/supplementary/special exam application, graduation
Student Counselling & Career Resource Centre Location	Lower Library Complex, Steve Biko Campus, next to financial Aid Reception Desk (031)373 2266 Career Resource Centre (031)373 2571	Individual counselling, career guidance, study skills support
HIV/AIDS Centre	Front Desk (031) 373 2260	Emotional support
Health Clinic	Isolempilo(Gate 5, opposite Sports Centre) 031 373 2223 Ritson Campus(Next to Hotel School Restaurant) 031 373 6010	Primary health services, Family Planning, HIV Testing
Financial Aid Services and Scholarships	Lower Library Complex, Steve Biko Campus 031 373 2553 applications@nsfas.org.za NSFAS call centre on 0860 067 327	Application for financial support
Student Residence	Indira Khiramen Tel: 031 373 2494 Email: indirak@dut.ac.za	Application for residence whilst studying at DUT

9. Quality assurance

Learners are encouraged to provide feedback on all aspects of this course and are advised to complete the LEQ and SEQ administered at the end of the course. More importantly, learners that experience any difficulties or have any complaints about this course should feel free to contact the class representative or the lecturer.

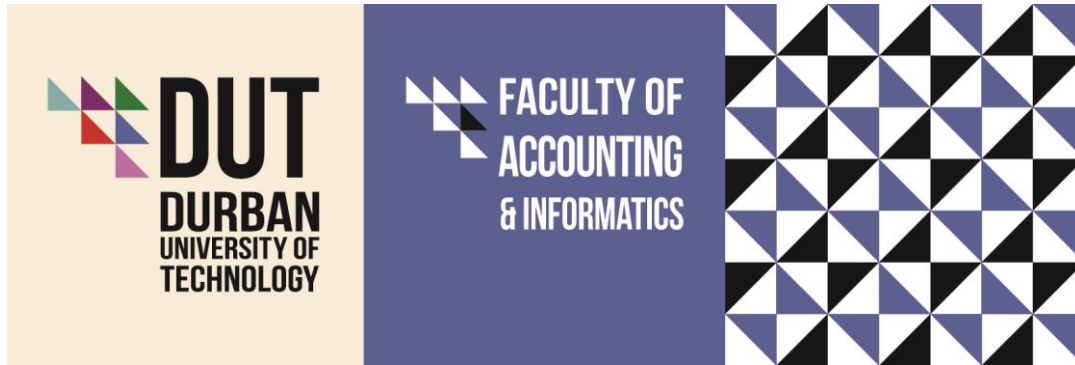
10. Work Integrated Learning (WIL), industry, community, and occupation-related information

Please note that WIL is not applicable to this module. Practices (e.g., research, team work, etc.) and tools used relevant to the Information Technology industry in general, and software development in particular, are emphasised.

11. Changes to this document

Please note that the content of this study guide is subject to change. Any changes thereof will be published on Moodle.

Appendix A – Plagiarism declaration



PLAGIARISM DECLARATION

(TO BE SUBMITTED BY EACH STUDENT FOR EVERY SUBMISSION)

Increment _____

Group #: _____

DECLARATION

1. I know and understand that plagiarism is using another person's work and pretending it is one's own, which is wrong.
2. This assignment is my own work.
3. I have appropriately referenced the work of other people I have used.
4. I have not allowed, and will not allow, anyone to copy my work with the intention of passing it off as his or her own work.

Signature

Date

Name (in capital letters)

Student Number